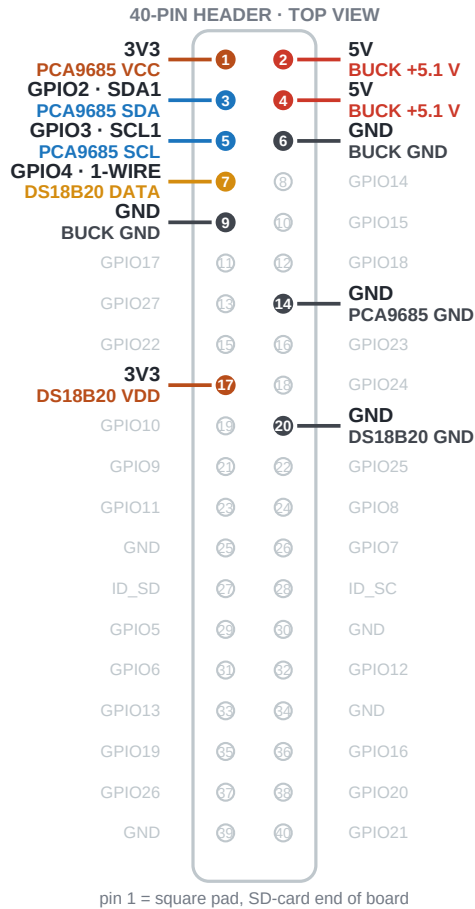




POWER — HEADER INJECTION

Supply	5.1 V at the header · 5 A min · protected rail
5 V in	pins 2 + 4 (both — split the current)
GND	pins 6 + 9 (two returns)
Wire	18 AWG · < 15 cm buck-to-header
Rule	never plug USB-C while the rail is live

I2C — PCA9685 @ 0x40

SDA	pin 3 (GPIO2)
SCL	pin 5 (GPIO3)
VCC logic	pin 1 (3V3)
GND	pin 14
V+	EMPTY — open-drain mode, no supply

1-WIRE — DS18B20 ON GPIO4

DATA (yel)	pin 7 (GPIO4)
VDD (red)	pin 17 (3V3) — powered, not parasite
GND (blk)	pin 20
Pull-up	4.7 k ohm DATA -> 3V3, at the Pi end
Overlay	dtoverlay=w1-gpio-pi5,gpiopin=4 (set)

DIM RAIL — CH0 -> XLG-50-AB x2

PCA CH0	-> both driver DIM- lines (parallel)
Pull-up	10 k ohm CH0 line -> +10 V rail
Rail GND	common with PCA GND + buck GND

COMMON GROUND — ONE NET

Pi GND = buck GND = PCA9685 GND =

10 V dim-rail GND = DS18B20 GND

Floating dim-rail ground = erratic dimming. #1 miss.

VERIFY AFTER WIRING

```
i2cdetect -y 1      -> 0x40 (+0x70 all-call = normal)
ls /sys/bus/w1/devices/ -> 28-*
```